

Experiment - 1

AIM:

To design an Entity - Relationship (ER) diagram for an Indian e-commerce Platform consisting of Customer, Product, Order, Order Item, Seller, Category, Payment, Delivery and Address, and to identify primary keys, composite and multi-valued attributes, weak entities, specialization generalization, and Participation constraints.

* ER Diagram Components

* Entities:

- | | | |
|-------------|---------------|-------------|
| 1. Customer | 4. Order Item | 7. Payment |
| 2. Product | 5. Seller | 8. Delivery |
| 3. Order | 6. Category | 9. Address |

Primary keys:

Customer → Customer_ID

Product → Product_ID

Order → Order_ID

Order Item → (Order_ID, Item No) [Composite Primary key]

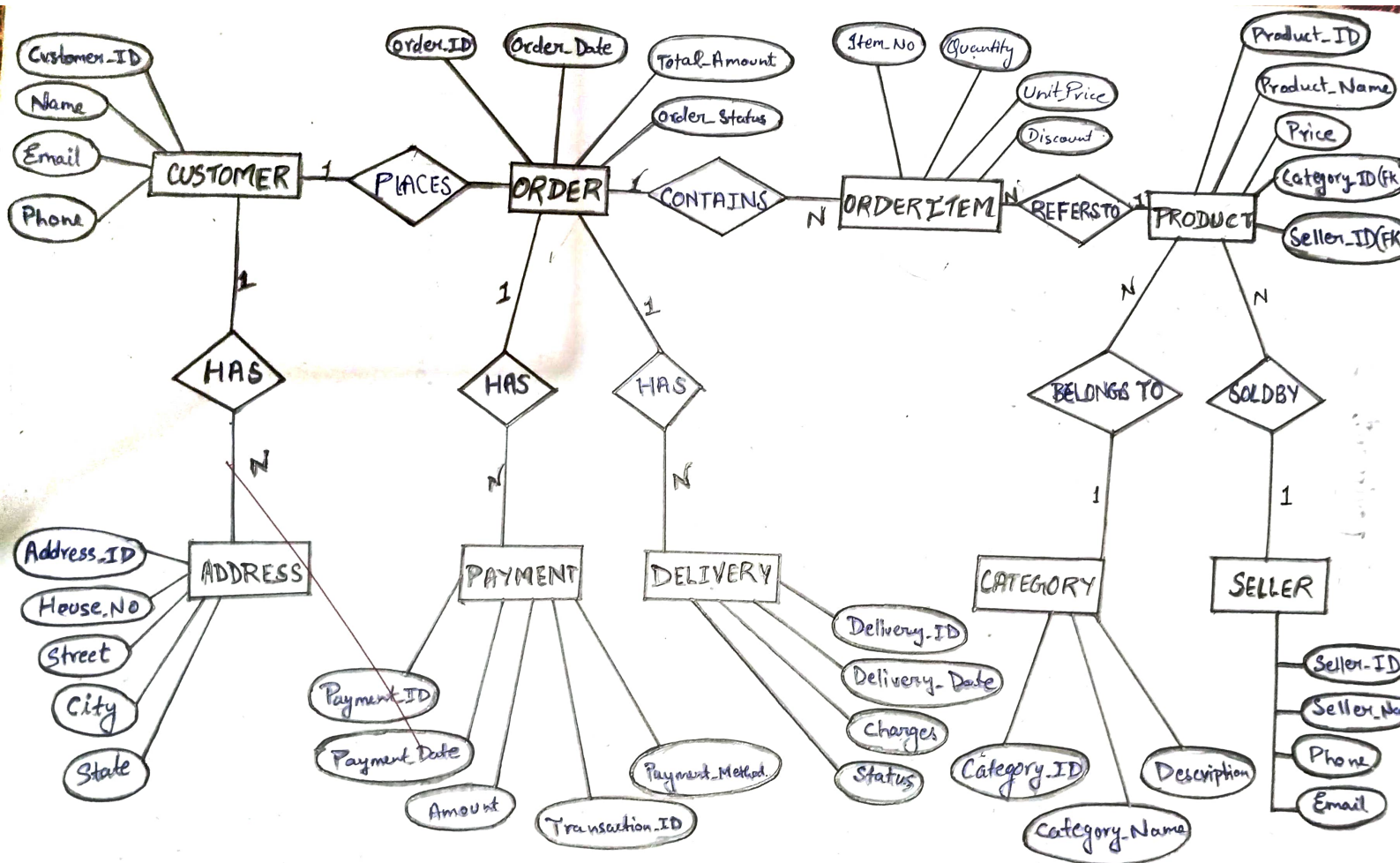
Seller → Seller_ID

Category → Category_ID

Payment → Payment_ID

Delivery → Delivery_ID

Address → Address_ID



* Important Attributes

1. Customer

- Customer-ID
- Name
- Phone

2. Product

- Product-ID
- Product-Name
- Price

3. Order

- Order-ID
- Order-Date
- Total-Amount
- Order-Status

4. Order Item

- Item-No
- Quantity
- Unit-Price
- Discount

5. Seller

- Seller-ID
- Seller-ID Name
- Phone

6. Category

- Category-ID
- Category-Name
- Description

7. Payment

- Payment-ID
- Payment-Date
- Amount
- Transaction-ID
- Payment-Method

8. Delivery

- Delivery-ID
- Delivery-Date
- Status
- charges

9. Address

- Address-ID
- House-No
- Street
- City
- State
- Pincode

* Composite Attribute

- Customer_Name can be represented as:

Name $\rightarrow$ First_Name, Middle_Name, Last_Name.

- Address can be represented as:

Address $\rightarrow$ House-No, Street, City, State, Pincode.

* Multi-Valued Attribute

Customer.Phone is a multivalued attribute because a customer may have more than one phone number.

* Weak Entity

OrderEntityItem is a weak entity because it can not exist independently without an order.

* Specialization of Product

Product is specialized using an ISA relationship:

Product $\rightarrow$ Electronics, Clothing
Electronics $\rightarrow$ Mobile, Laptop

Eg:

- Electronics : Brand, Warranty-Period
- Clothing : Size, Fabric, Colour

* Relationships and Cardinality

1. Customer Places Order $\rightarrow 1:N$
2. Customer has Address $\rightarrow 1:N$
3. Order Contains OrderItem $\rightarrow 1:N$
4. Product appears in OrderItem $\rightarrow 1:N$
5. Seller Sells Product $\rightarrow 1:N$

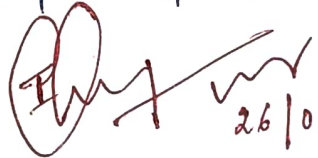
6. Category contains Products $\rightarrow 1:N$

7. Order has Payment $\rightarrow 1:1$

8. Order has Delivery $\rightarrow 1:1$

↳ Participation Constraints -

- Customer - Order : Order has total Participation
- Order - order item : Both order and Order Item have total Participation
- Product - Order Item : Order Item has total Participation
- Seller - Product : Product has total Participation
- Category - Product : Product has total Participation
- Order - Payment : Payment has total participation
- Order - Delivery : Delivery has total participation

 26/08/26